Veesam Aakanksha

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PROFESSIONAL SUMMARY

AI/ML Engineer with 5 years of experience in machine learning model development, MLOps, and AI automation across finance, healthcare, and e-commerce. Strong expertise in ML model deployment (AWS SageMaker, Azure ML), cloud-based AI solutions, and automating data pipelines. Skilled in fraud detection, risk modeling, anomaly detection, NLP, and business intelligence automation (Power Automate, Power Apps, Power BI).

Proven ability to optimize ML models, automate workflows, and integrate AI with business applications. Experience in explainable AI (SHAP, LIME) for regulatory compliance and scaling AI microservices using FastAPI and Kubernetes. Passionate about building high-performance, cloud-native AI solutions that enhance operational efficiency and decision-making.

Additionally, I have developed AI-powered document processing solutions using OCR and NLP, reducing manual effort and improving document classification and extraction accuracy. Passionate about bridging AI and real-world impact, I focus on high-performance AI deployments that drive business innovation, automation, and operational efficiency.

TECHNICAL SKILLS

Programming Languages: Python, R, Java, Scala, SQL, MATLAB | **Big Data Technologies**: Hadoop, Spark, MapReduce, HDFS | **Databases**: SQL Server, PostgreSQL, MongoDB, Cassandra, Redis, MySQL | **Natural Language Processing (NLP)**: Text Analytics, Lemmatization, Tokenization, Speech Analytics, NER, Sentiment Analysis| **Machine Learning & AI**: Linear & Logistic Regression, Naive Bayes, Random Forests, Decision Trees, SVM, Gradient Boosting (XGBoost, LightGBM, CatBoost), Autoencoders, Reinforcement Learning, SHAP & LIME | **Model Evaluation & Optimization**: Cross-Validation, Bayesian Optimization, Regularization, ANOVA, Hypothesis Testing | **Libraries & Frameworks**: NumPy, SciPy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost | **Cloud & DevOps**: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Terraform, Jenkins | **Data Visualization & Reporting**: Tableau, Power BI, Matplotlib, Seaborn, Plotly | **Development Tools**: JupyterLab, Visual Studio Code, Spyder, Git, Apache Airflow, MLflow

WORK EXPERIENCE

AI & ML Engineer | Amex | Texas **Jan 2024 - Present**

- Currently leading the development of a Credit Risk Assessment Engine, migrating models to cloud-native deployments using AWS SageMaker and Azure ML to reduce training time by 35% and inference latency by 40%.
- Led development of an end-to-end MLOps pipeline using MLflow, Docker, and Jenkins and Terraform for model versioning, packaging, and automated deployment to SageMaker endpoints.
- Built and deployed risk scoring models using Logistic Regression, Random Forests, and Deep Neural Networks with TensorFlow and Scikit-learn, improving credit approval accuracy by 22% across key customer segments.
- Automated compliance workflows using Apache Airflow and Power Automate, integrating ANOVA-based statistical checks and SHAP-driven transparency for regulatory audits.
- Integrated real-time risk scores into a Spark-based data pipeline, leveraging Kafka and Redis for low-latency event processing across millions of applications daily.
- Developed explainability dashboards using SHAP and LIME with Power BI and Plotly, enabling risk teams to audit model behavior and justify approvals or rejections in customer journeys.
- Implemented automated drift detection and retraining using Apache Airflow and Spark Streaming, improving SLA compliance by 28%.
- Optimized model tuning with cross-validation and Bayesian Optimization using Scikit-learn, boosting recall in underperforming credit segments by 17%.
- Conducted drift detection and automatic retraining triggers via Spark Streaming, ensuring continuous model performance on streaming data from HDFS-backed pipelines.

AI Engineer | Deloitte | India **Mar 2021 - Dec 2022**

- Developed predictive analytics models (Logistic Regression, LightGBM) to enhance business forecasting accuracy and inform strategic decision-making.
- Designed Python-based ETL pipelines, automating data preprocessing workflows & reducing manual intervention by 35%.
- Developed a deep learning model using CNNs to classify medical X-ray images, achieving 85% accuracy in pneumonia detection, aiding in early diagnosis.
- Implemented AI-based predictive analytics for patient readmission risk, using XGBoost, improving hospital resource allocation efficiency by 30%.
- Conducted A/B testing on AI-driven features, validating model performance and ensuring statistically significant improvements in business impact.
- Optimized SQL-based data pipelines & feature engineering workflows, reducing ML model training time by 25%, while designing analytical dashboards in Power BI & Tableau to enhance data-driven decision-making.
- Developed an AI-powered OCR pipeline using Tesseract & OpenCV, automating invoice text extraction. Integrated Power Automate for automated document routing and approval workflows, reducing processing time by 12000 man hours.
- Engineered an ML-based employee attrition prediction system, leveraging XGBoost and feature engineering to enhance HR retention strategies by 15%.
- Supported AI-driven chatbot development, integrating Named Entity Recognition (NER) models for enhanced customer and HR automation, improving employee engagement and support efficiency.

- Built anomaly detection pipelines using Scikit-learn & Pandas, improving data quality & reducing inconsistencies in client datasets.
- Collaborated with cross-functional teams to translate business problems into AI-driven solutions, ensuring alignment between ML models and business objectives.

Data Analyst | IHS Markit | India **Dec 2019 - Feb 2021**

- Developed 25+ automated data analytics pipelines, integrating Python-based ETL workflows and machine learning models, improving data visibility by 35%.
- Built 40+ Power BI, and Tableau dashboards, providing real-time insights for proactive financial decision-making.
- Optimized SQL-based feature engineering pipelines, improving data retrieval speeds by 20% & model training efficiency.
- Implemented anomaly detection models using unsupervised learning (Isolation Forest, DBSCAN) to identify data inconsistencies.
- Utilized AWS ECS, Lambda, Batch and Opensearch for backend development and cloud deployment and wrote customized CloudFormation scripts for better performance.
- Automated MLOps workflows, integrating CI/CD pipelines (Jenkins, GitHub Actions, Python automation), reducing MTTR by 20%.
- Created and maintained various DevOps related tools for the team such as provisioning scripts, deployment tools and staged virtual environments using Docker.
- Enhanced cloud cost efficiency by analyzing resource utilization patterns, reducing unnecessary compute costs by 15%.
- Developed automated data validation pipelines, reducing manual error-checking efforts by 30%, ensuring high-quality ML input data.

ACADEMIC PROJECTS EXPERIENCE

FairShare | Expense dividing System

- Designed an intelligent expense management system using Flask and MySQL, integrating predictive analytics and Deep Learning (LSTMs) for personalized budget recommendations, achieving 92% accuracy in expenditure forecasting.
- Built a machine learning-based recommendation engine, combining collaborative filtering and classification models to analyze user behavior and enhance purchase prediction accuracy.
- Automated bill processing and expense tracking with OCR and NLP-based document parsing, reducing manual entry time by 80% and achieving 98% accuracy in transaction detail extraction.
- Developed a personalized expense categorization model using NLP and clustering techniques, enabling users to gain deeper insights into their spending habits through automated transaction labeling.

Budget Buddy | Expense Tracker Application

- Developed an AI-driven expense categorization system using Logistic Regression & XGBoost, automating 80% of manual transaction tagging.
- Integrated receipt scanning APIs to extract transaction details, utilizing OCR & NLP-based entity recognition to improve data accuracy by 30%.
- Streamlined placements through containerization with Docker & optimized build times using Vite, improving overall performance

NOTED APP | AI-Powered Meeting Notes Summarizerll

- Developed an AI-driven meeting notes summarization system using DistilBERT & T5, automatically generating concise summaries from long meeting transcripts, improving documentation efficiency.
- Integrated Azure Speech-to-Text API, enabling real-time transcription of meetings, converting spoken words into structured text with 85%+ accuracy.
- Deployed the model as a FastAPI microservice, storing structured summaries in PostgreSQL for quick retrieval, reducing manual note-taking efforts in enterprise environments.

Sentiment Analysis for Movie Reviews using LSTM

- Trained an LSTM-based sentiment analysis model with pretrained BERT embeddings, achieving 94% accuracy on Stanford dataset, improving sentiment classification over traditional word vector models.
- Implemented text preprocessing pipelines including tokenization, lemmatization, stop-word removal, and embedding layers, improving model generalization across diverse movie review datasets.
- Deployed the model using TensorFlow Serving with REST API endpoints, optimizing inference with FastAPI and Docker, reducing latency to under 70ms per request, supporting 100+ concurrent users.

EDUCATION

Master of Science in Computer Science, University of Texas at Arlington | Arlington, TX | **GPA**: 3.86/4.0 **Dec 2024**
Bachelor in Information technology, Vasavi College of Engineering | India | **GPA**: 3.60/4.0 **Jun 2022**

CERTIFICATIONS

- **AWS** Certified solutions architect
- **Machine Learning:** Regression and classification
- **Oracle** Cloud Infrastructure AI Foundations All Users
- **Oracle** Cloud Infrastructure Data Foundations Associate
- **Power Apps -** App in a Day, Pragmatic Works
- **Networking:** The Bits and Bytes of Computer Networking